ENVS 193DS Infographic

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Set up

```
# loaded in packages
library(tidyverse)
library(janitor)
library(here)
library(webr)
library(ggimage)
library(showtext)

library(dplyr)
library(ggplot2)

# read in data and saved as object called `rxntime`
rxntime   <- readr::read_csv(here::here("data","new-data.csv"))

# read in my data and saved as object called 'pie'
pie <- readr::read_csv(here::here("data","pie-data.csv"))

# imported fonts
# 1st argument: google name,
# 2nd name: font name in R
font_add_google('Exo 2', 'exo2', 700)
font_add_google('Exo 2', 'exo')

# enabled showtext font rendering
showtext_auto()
```

Part 3. Visualizing data

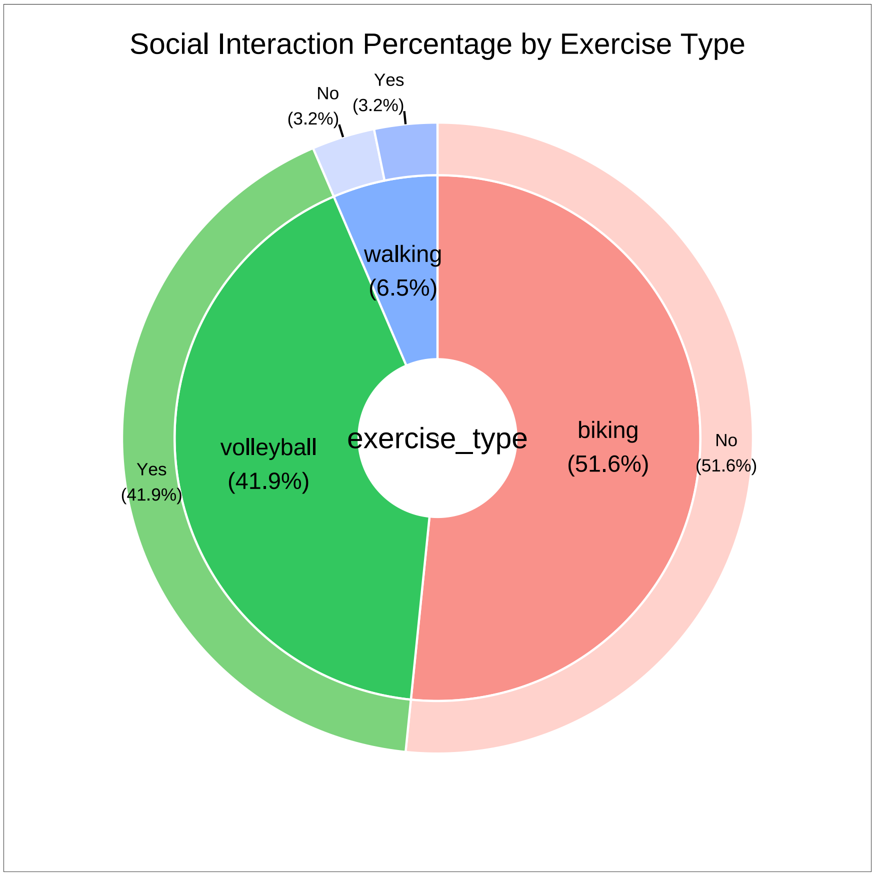
Pie chart

Cleaning data

```
# created a new object from pie
pie_clean <- pie |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, social_interaction, count) |>
  # changed existing abbreviations
  mutate(social_interaction = case_when(
    social_interaction == "y" ~ "Yes",
    social_interaction == "n" ~ "No"
  )) |>
  # removed missing values
  drop_na()
```

Visualization

```
# save plot as an object <- created a pie donut chart with pie_clean df
piechart <- PieDonut(data = pie_clean,
         # pie slices by exercise type, outlined by social interaction
         aes(exercise_type, social_interaction, count = count),
         r0 = 0.3, # inner ring thickness
         r1 = 1, # outer ring thickness
         ratioByGroup = FALSE, # outer circle of the pie adds up to 100%
         title = "Social Interaction Percentage by Exercise Type") # title
```




```
# PNG export
ggsave("C:/github/ENVS-193DS_spring-2026_infographic/exports/pie.png",
       plot = get_last_plot(),
       device = ragg::agg_png,
       bg = "transparent",
       width = 6,
       height = 6,
       units = "in",
       dpi = 300)
```

Scatter plot

Cleaning data

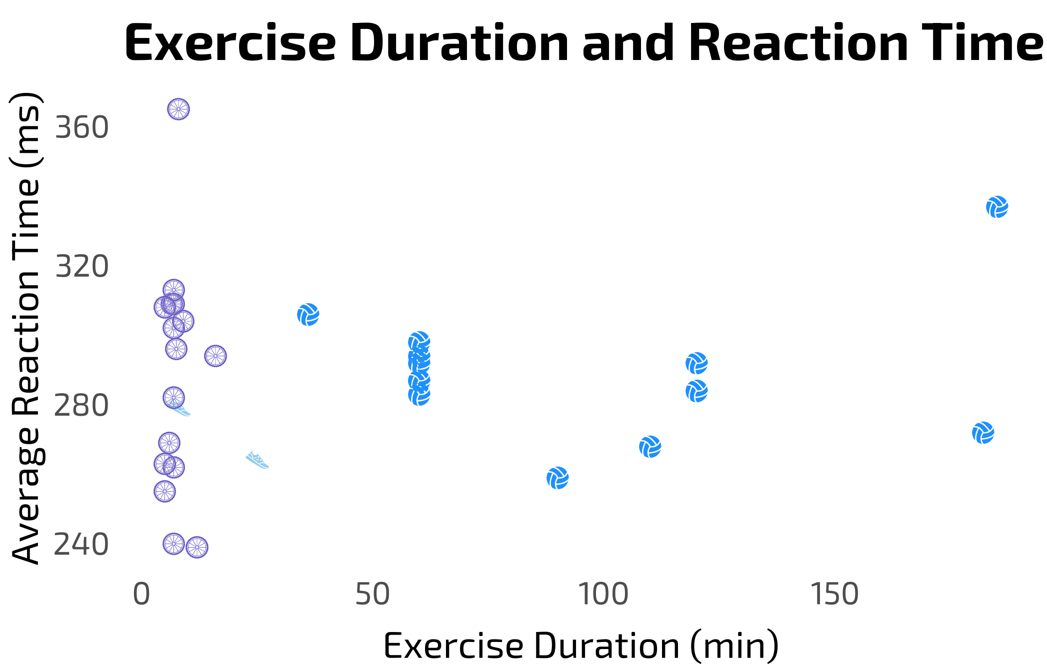
```r
# created a new object from rxntime
rxntime_clean <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, exercise_duration_min, reaction_time_avg_ms) |>
  # removed missing values
  drop_na()
```

Visualization

```r
# created image map by exercise type
img_map <- c(
  biking = "C:/github/ENVS-193DS_spring-2026_infographic/images/biking.png",
  volleyball = "C:/github/ENVS-193DS_spring-2026_infographic/images/vb.png",
  walking = "C:/github/ENVS-193DS_spring-2026_infographic/images/walking.png"
)

# new object from clean data frame
rxntime_scatter <- rxntime_clean |>
  # mutated data frame for the addition of icons
  mutate(exercise_type = as.character(exercise_type),
         image = img_map[exercise_type])
```

```r
# called showtext
showtext_auto()

# saved plot as an object <- base layer: ggplot
scatterplot <- ggplot(rxntime_scatter,
            # x-axis: exercise duration in minutes
            # y-axis: avg reaction time in milliseconds
            aes(x = exercise_duration_min,
                y = reaction_time_avg_ms)) +
  # first layer: data points represented by custom icons
  geom_image(aes(
    image = image), # icon placement
    size = 0.05, # icon size
    asp = 1) + # square aspect ratio
  # custom theme
  theme_minimal() +
```

```
  theme(
    legend.position = "none", # removed legend
    panel.background = element_rect(fill = NA, # removed fill and color
                                    color = NA),
    panel.grid = element_blank(), # removed panel grid
    axis.ticks = element_blank(), # removed axis ticks
    plot.background = element_rect(fill = NA, # removed fill and color
                                  color = NA),
    plot.title = element_text(family = "exo2", # title custom font
                             size = 20),
    axis.title = element_text(family = "exo", # axis title font
                             size = 14),
    axis.title.x = element_text(margin = margin(t = 8)), # text spacing
    axis.text = element_text(family = "exo", size = 12)) + # tick label
  # relabeled x- and y- axes, adding title
  labs(title = "Exercise Duration and Reaction Time",
       x = "Exercise Duration (min)",
       y = "Average Reaction Time (ms)")

# view figure in plots
print(scatterplot)
```



```r
# PNG export
ggsave("C:/github/ENVS-193DS_spring-2026_infographic/exports/scatterplot.png",
       plot = get_last_plot(), # selected plot to save
       device = ragg::agg_png, # supports alpha
       bg = "transparent", # transparent background
       width = 4, # adjusted width
       height = 3, # adjusted height
       units = "in", # units of plot measurements
       dpi = 300) # resolution
```

Box / jitter plot

Cleaning data

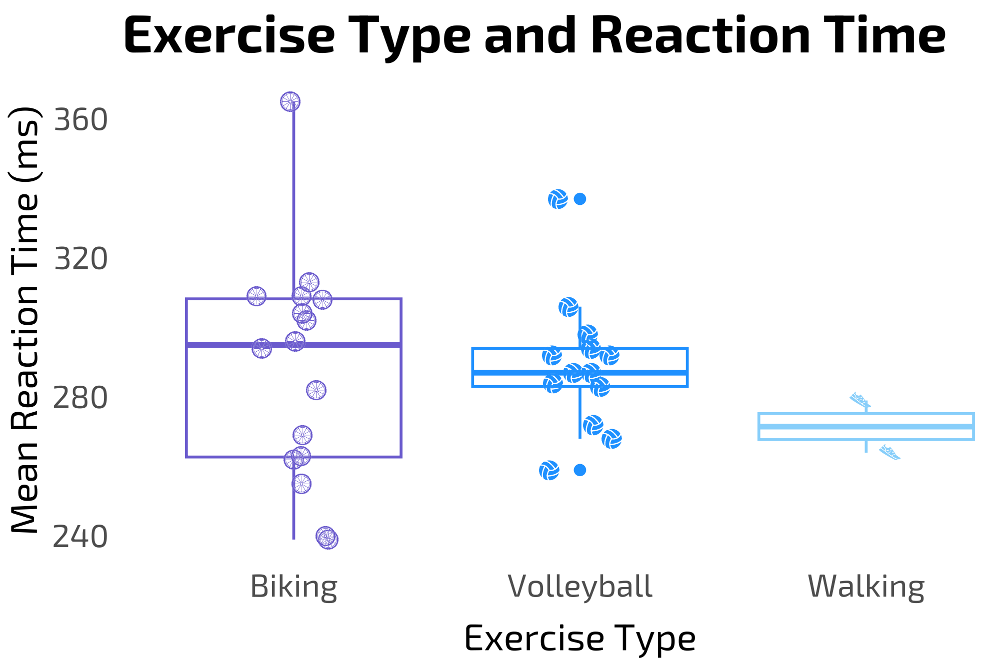
```r
# created a new object from rxntime
rxntime_box <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, reaction_time_avg_ms) |>
  # created a new column for images in df
  mutate(image = img_map[exercise_type]) |>
  # capitalized existing abbreviations
  mutate(exercise_type = case_when(
    exercise_type == "volleyball" ~ "Volleyball",
    exercise_type == "biking"     ~ "Biking",
    exercise_type == "walking"    ~ "Walking",
    # assigned exercise types to images
    TRUE ~ as.character(exercise_type)
  )) |>
  # removed missing values
  drop_na()
```

Visualization

```r
# called showtext
showtext_auto()
```

```r
# save plot as an object <- base layer: ggplot
boxplot <- ggplot(data = rxntime_box,
       # x-axis: exercise type
       # y-axis: mean reaction time
       # color by exercise type
       mapping = aes(x = exercise_type,
                     y = reaction_time_avg_ms,
                     color = exercise_type)) +
  # first layer: boxplot
  geom_boxplot() +
  # second layer: icons and jitter
  geom_image(
    aes(image = image),
    position = position_jitter(width = 0.15,
                               height = 0),
    size = 0.045, # icon size
    asp = 1) + # icon aspect ratio
  # added title and relabeled x- and y-axes
  labs(title = "Exercise Type and Reaction Time",
       x = "Exercise Type",
       y = "Mean Reaction Time (ms)") +
  # custom theme
  theme_minimal() +
  # removed legend
  theme(legend.position = "none", # removed legend
        panel.background = element_rect(fill = NA, # removed fill and color
                                             color = NA),
        panel.grid = element_blank(), # removed panel grid
        axis.ticks = element_blank(), # removed axis ticks
        plot.background = element_rect(fill = NA, # removed fill and color
                                           color = NA),
        plot.title = element_text(family = "exo2", # title custom font
                                      size = 20),
        axis.title = element_text(family = "exo", # axis title font
                                      size = 14),
        axis.title.x = element_text(margin = margin(t = 8)), # text spacing
        axis.text = element_text(family = "exo", size = 12)) + # tick label font
  # custom colors
  scale_color_manual(values = c("Biking" = "slateblue",
                                "Volleyball" = "dodgerblue",
                                "Walking" = "lightskyblue")) +
  # removed grid and axis lines
```

```
  theme(panel.grid = element_blank(),
        axis.line = element_blank())

# view figure in plots
print(boxplot)
```




```
# PNG export
ggsave("C:/github/ENVS-193DS_spring-2026_infographic/exports/box.png",
       plot = get_last_plot(), # selected plot to save
       device = ragg::agg_png, # supports alpha
       bg = "transparent", # transparent background
       width = 4, # adjusted width
       height = 3, # adjusted height
       units = "in", # units of plot measurements
       dpi = 300) # resolution
```

Part 4. Write about your process

A classmate on Friday used birds to represent raw data points.

e.

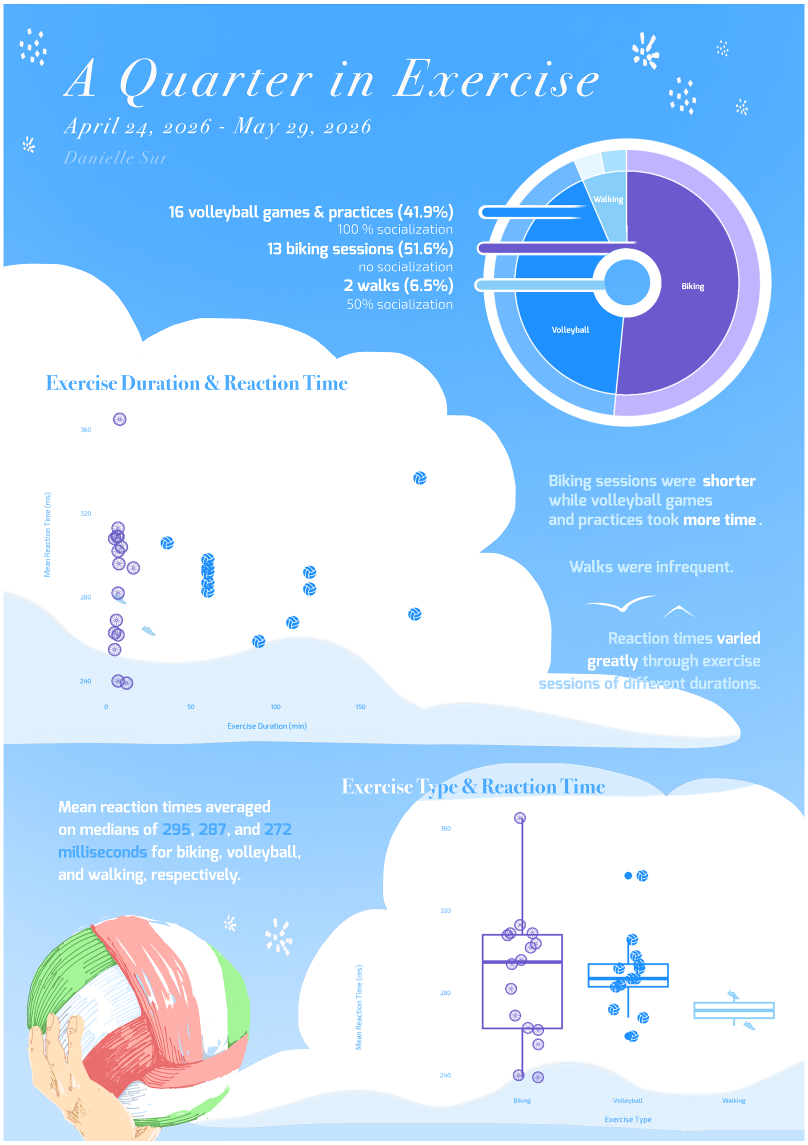

A Quarter in Exercise
April 24, 2026 - May 29, 2026
Danielle Sui
16 volleyball games & practices (41.9%)
100 % socialization
13 biking sessions (51.6%)
no socialization
2 walks (6.5%)
50% socialization
Walking
Biking
Volleyball
Exercise Duration & Reaction Time
Mean Reaction Time (ms)
Exercise Duration (min)
Biking sessions were shorter while volleyball games and practices took more time.
Walks were infrequent.
Reaction times varied greatly through exercise sessions of different durations.
Exercise Type & Reaction Time
Mean reaction times averaged on medians of 295, 287, and 272 milliseconds for biking, volleyball, and walking, respectively.
Exercise Type

Figure 1